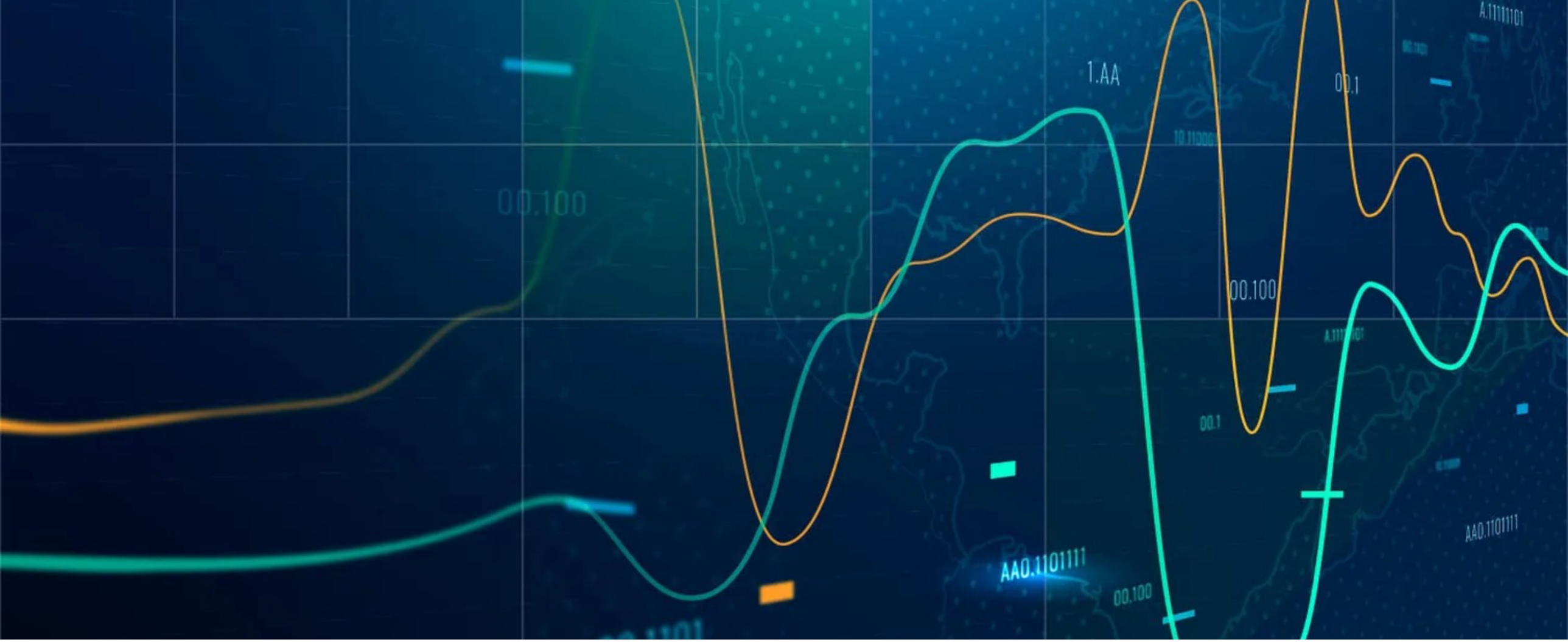

2110446 - Data Science and Data Engineering

# Timeseries Datastore - Prometheus

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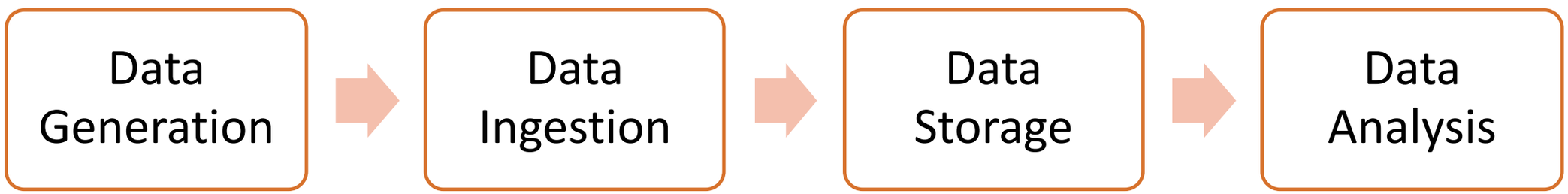
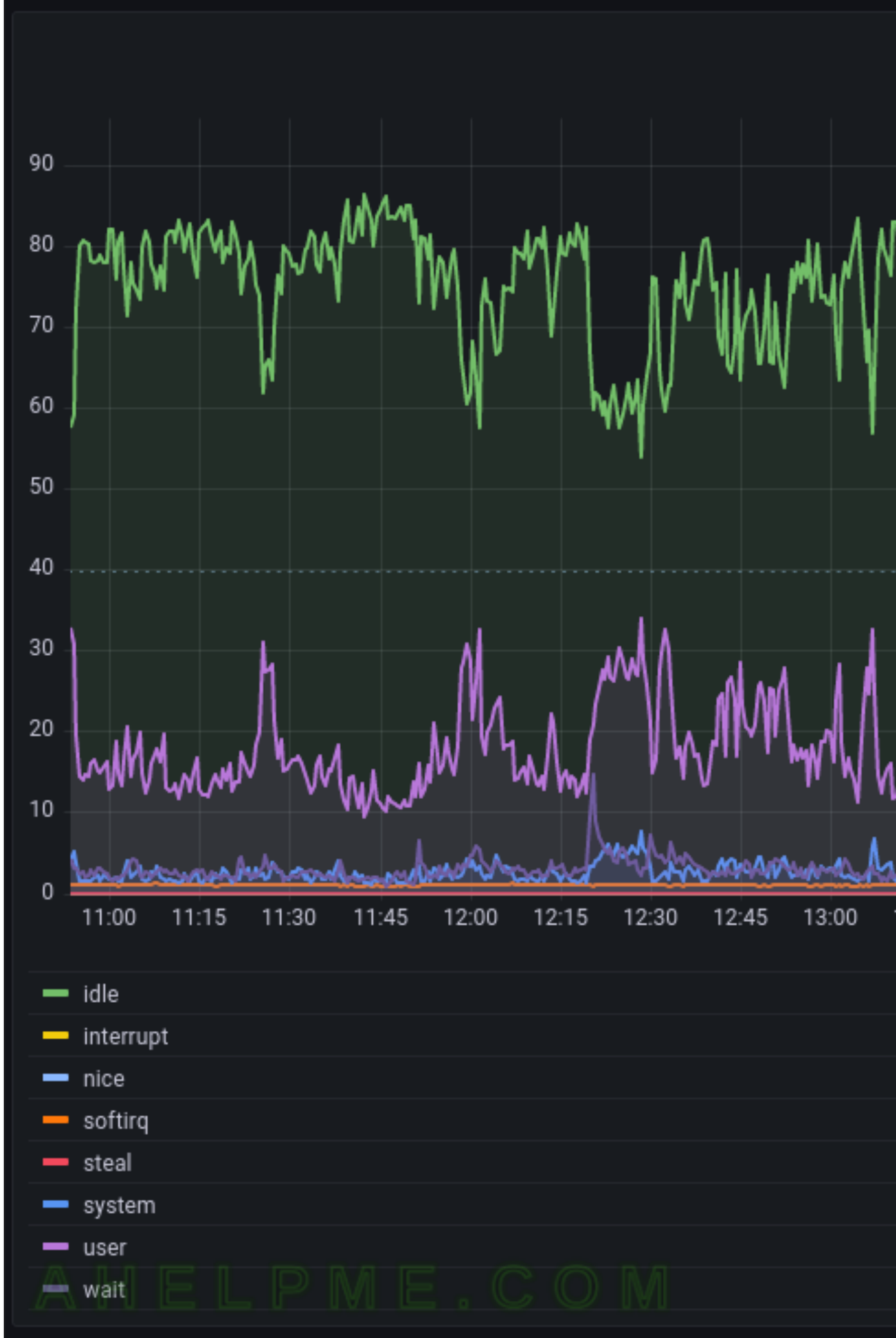
# Data Lifecycle

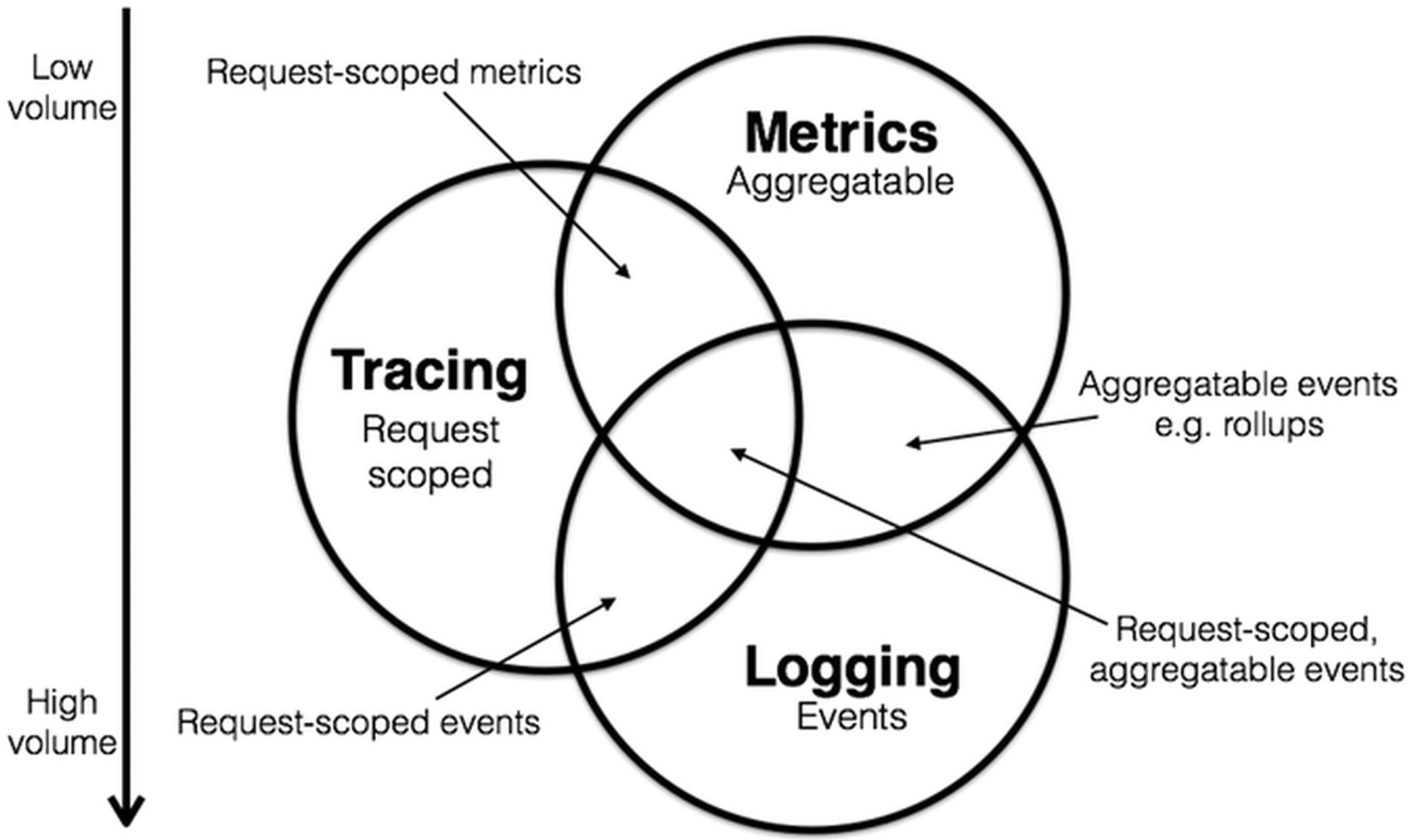


# TSDB: Time Series Database

- A sequence of data points recorded at specific time intervals
- The Timestamp is the primary key and the most critical dimension
- Example:
  - CPU workloads from servers
  - Heart rate monitor from ICU
  - Stock market quotes
  - Air quality sensors
- **Time series datastore is the heart and soul of modern observability**



# Three Pillars of Observability




Source: https://medium.com/kbtg-life/observability-pt-1-ทำความรู้จักกับ-observability-กันเถอะ-5b2561bc4365

# Time Series Workload Characteristics

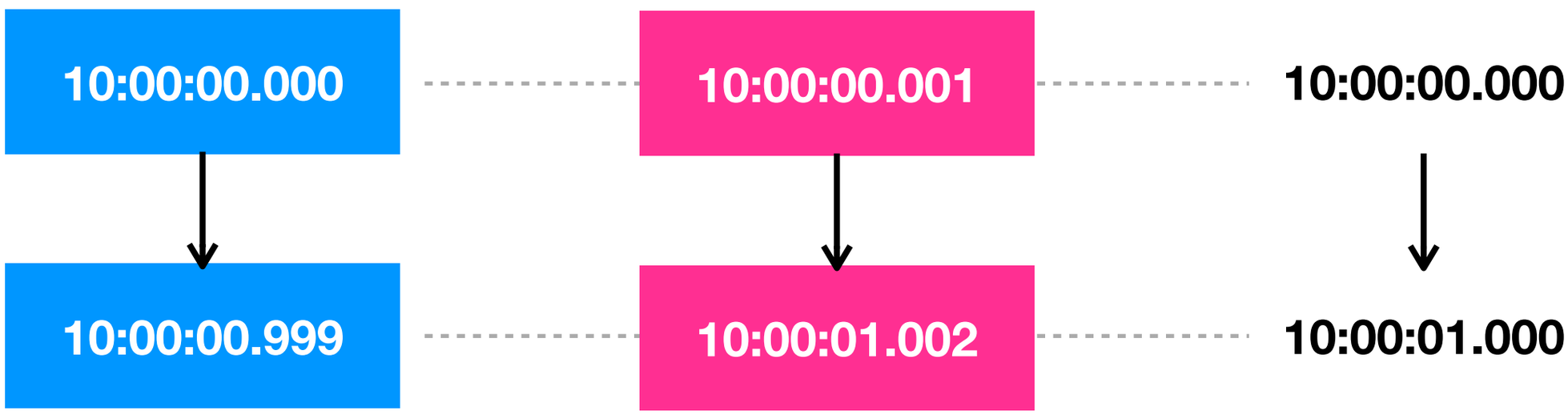
- **High Ingestion**: Constant streams of appends
- **Immutable**: Historical data is rarely changed
- **Temporal Locality**: Recent data is queried 95% of the time
- **High Compression Ratio**: Several techniques are designed for time series data e.g. delta, delta-of-delta, etc.
- **Recent data is most valuable**: Downsampling to reduce storage footprint for long-term retention

# Lifecycle of Time Series Data

1. **Ingestion**: High-velocity data streams
2. **Storage**: Efficient compression for long-term history
3. **Querying**: Range-based lookups, fix-window aggregations, sliding-window operations
4. **Retention**: Deleting or downsampling old data for more efficient long-term retention

# NoSQL on Time Series Data Limitations

- **Index Bloat**: Indexing timestamps in a Document DB causes index sizes to explode, often exceeding the actual data size
- **Retention Pain**: Deleting billions of documents in a NoSQL cluster creates massive I/O overhead and table locks
- **Alignment**: TSDBs handle "sample alignment" (moving data into consistent time buckets e.g. every 15-min) natively; NoSQL requires app logic



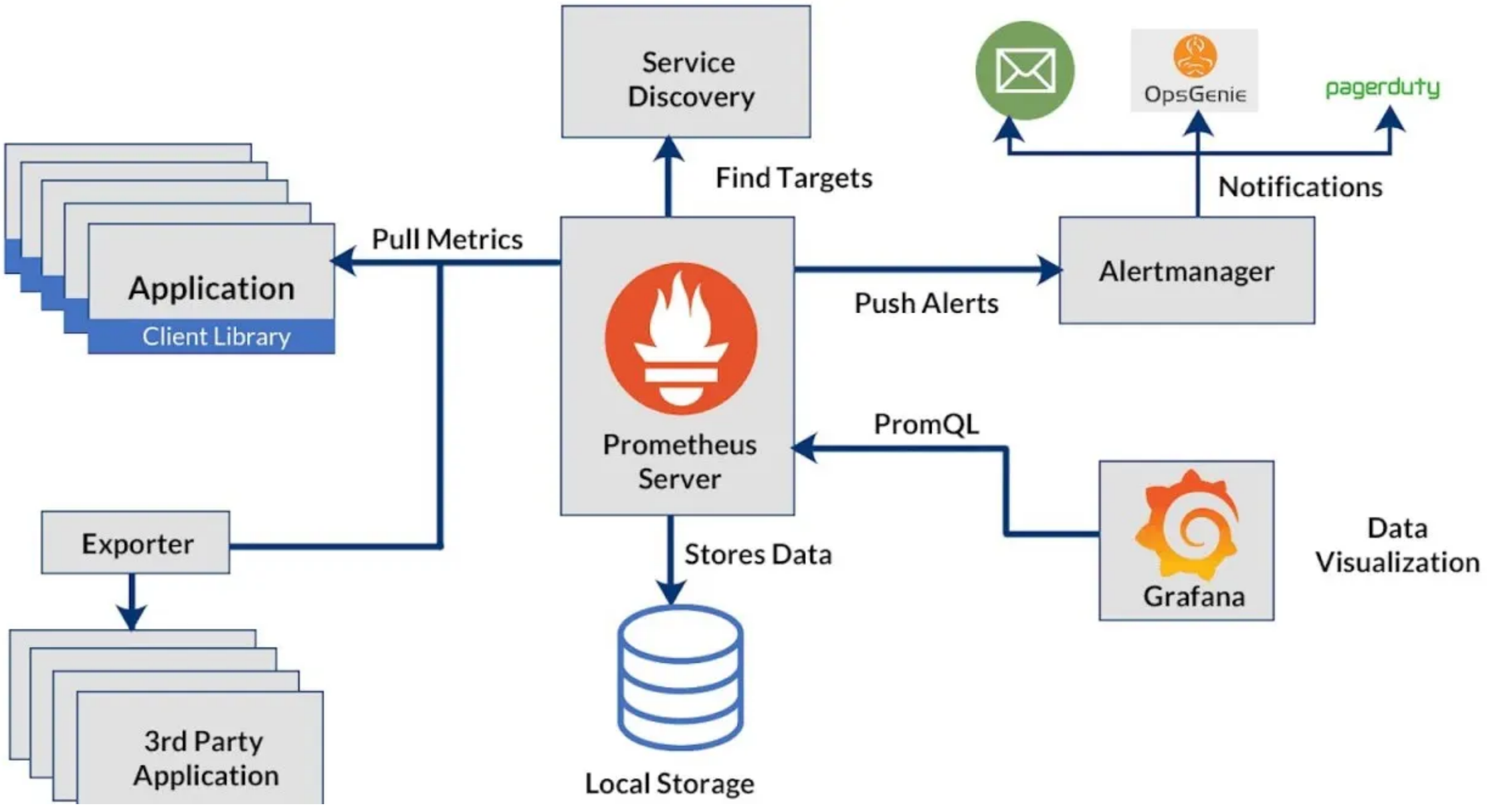
# Data Stores and Time Series Data

| Capability | Relational (RDBMS) | NoSQL (Document) | TSDB (Prometheus/Influx) |
|---|---|---|---|
| Ingestion | Transaction-Heavy | Scalable but Generic | Optimized Appends |
| Storage Footprint Reduction | General (Inefficient) | Minimal | Delta-of-Delta (90% Red.) Downsampling |
| Aggregation | Complex Joins | Map-Reduce / Agg Pipes | Native Temporal Ops |

# TSDB with Prometheus

- An open-source systems monitoring and alerting toolkit
- Standard for Cloud-Native / Docker / Kubernetes
- Pulls (scrape) metrics from a client (target) over http
- Places the data into its time series database
- Query using its own DSL (PromQL)
- Have its own Web UI and integrate seamlessly with Grafana

# Prometheus Architecture




Source: https://medium.com/@munikanthtech/introduction-to-prometheus-architecture-setup-20034b885e2a

# Running Prometheus

- For this tutorial, we will run 3 containers:
  - prometheus (port 9090) - main TSDB with Web UI (via 9090)
  - node-exporter (port 9091) - an agent collecting hw/os metrics
  - grafana (port 3000) - visualization platform
- See “docker-compose.yml” and “prometheus/prometheus.yml” for more details

# PromQL

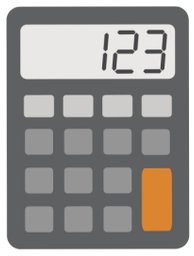
- Prometheus Query Language (PromQL) is a powerful and flexible language used to query and manipulate time-series metrics in Prometheus

- Key Concepts
  - Metrics and their types
  - Data types: instant vector, range vector, scalar, string
  - Expression: operators, functions

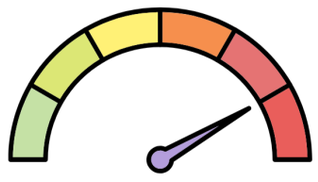
# Metrics

- Numeric representation of data measured over intervals of time
- Can be used to represent the behavior of a system over intervals of time in the present and future
- Anatomy of a metric
  - Metric Name: The "what" (e.g., node_cpu_seconds_total)
  - Labels: Dimensions for filtering (e.g., region=“asia-southeast-1a“, instance=“server-1”)
  - Timestamp: Unix epoch in milliseconds
  - Value: Standardized 64-bit floating point

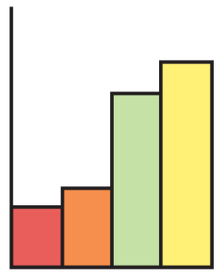
# Prometheus Metric Types



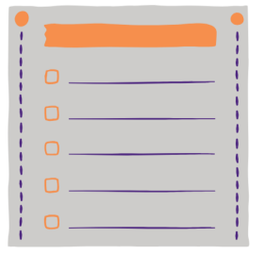
**Counter**



**Gauge**



**Histogram**



**Summary**

- **Counters** for tracking ever-increasing values, like the total number of exceptions thrown
- **Gauges** for measuring fluctuating values, such as current CPU usage
- **Histograms** for observing the distribution of values within predefined buckets (calculation on the server)
- **Summaries** for calculating quantiles (percentiles) of observed values (calculation on the client, sliding-window)

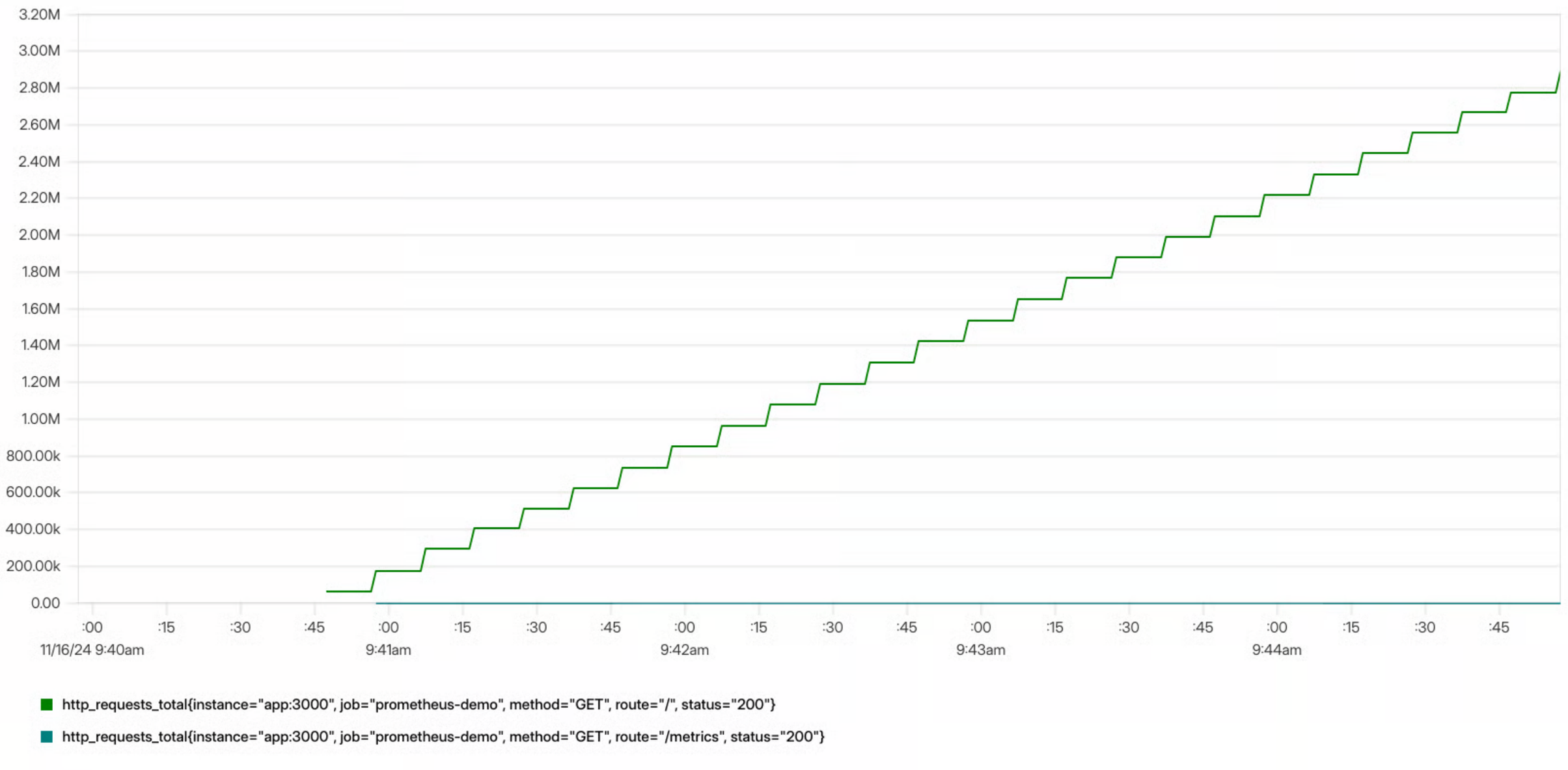
Prometheus Query Alerts Status

http_requests_total

Execute

Table Graph Explain

5m End time Medium res. Unstacked Stacked




Source: https://betterstack.com/community/guides/monitoring/prometheus-metrics-explained/

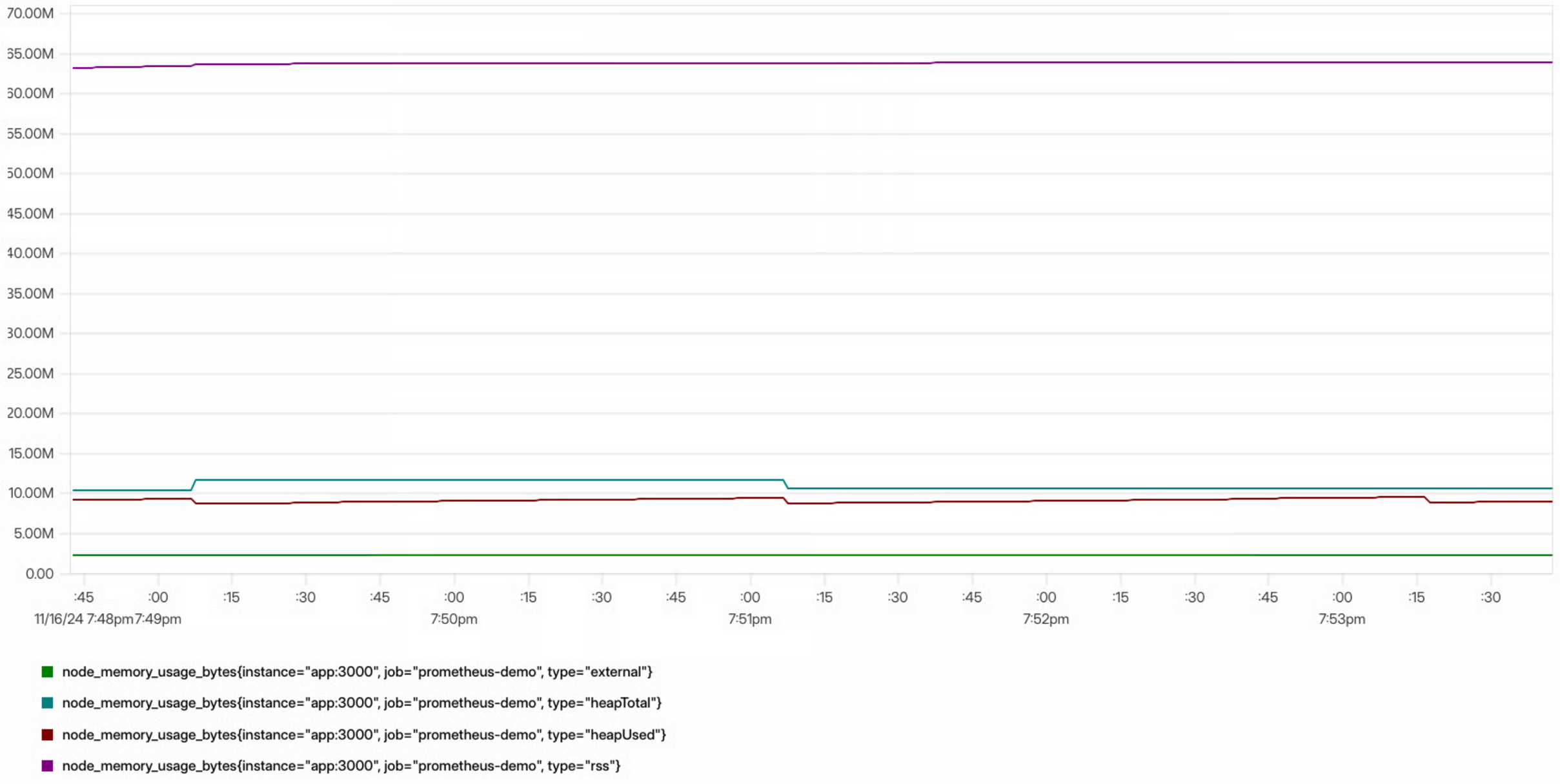
Prometheus Query Alerts Status

node_memory_usage_bytes

Execute

Table Graph Explain

5m End time Medium res. Unstacked Stacked




Source: https://betterstack.com/community/guides/monitoring/prometheus-metrics-explained/

Prometheus Query Alerts Status

http_request_duration_seconds_bucket

Execute

Table Graph Explain

Evaluation time

Load time: 9ms Result series: 12

| | |
|---|---|
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.005"} | 176 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.01"} | 176 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.025"} | 176 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.05"} | 176 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.1"} | 177 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.25"} | 2335 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="0.5"} | 5751 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="1.0"} | 5754 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="2.5"} | 5754 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="5.0"} | 5754 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="10.0"} | 5754 |
| http_request_duration_seconds_bucket{**instance**="app:3000", **job**="prometheus-demo", **le**="+Inf"} | 5754 |

Add query

Source: https://betterstack.com/community/guides/monitoring/prometheus-metrics-explained/

# Data Types

- In PromQL, an expression or sub-expression can evaluate to one of four types:
  - Instant vector - a set of time series containing a single sample for each time series, all sharing the same timestamp
  - Range vector - a set of time series containing a range of data points over time for each time series
  - Scalar - a simple numeric floating point value
  - String - a simple string value

# Instant Vector

- Containing the current value of that metric for each labeled instance at query time

```
<metric_name>{<label_selectors>}
```

- Examples:

```
node_cpu_seconds_total{mode="idle"}
node_cpu_seconds_total{mode="idle",cpu="0"}
```

- Each vector may contain many values due to the use of labels as each one creates its own time series

# Label Matching Operators

= Selects labels that are exactly equal to the provided string

!= Select labels that are not equal to the provided string

=~ Select labels that match a regular expression

!~ Select labels that do not match a regular expression

```
prometheus_http_requests_total{handler=~”/api/v1/.*"}
prometheus_http_requests_total{code!=“200"}
```

# Range Vector

- Containing the current value of that metric for each labeled instance at query time

```
<metric_name>{<label_selectors>}[duration]
```

- Examples:

```
node_cpu_seconds_total{mode=“idle”,cpu=“0"}[1h]
prometheus_http_requests_total{code=~"4.."}[5m]
```

# Instant vs. Range Vectors

- **Instant Vector**: Single value at 'now' (e.g. cpu_usage).

- **Range Vector**: Values over a duration (e.g. cpu_usage[5m] — if data is collected every 15-seconds, there are 5m x 4 datapoints/min = 20 datapoints)

- Note: You cannot graph range vectors directly; they must be aggregated

# Expressions

- An expression can be a simple metric name (node_cpu_seconds_total) or a complex combination of functions, operators, and selectors

- Operator: PromQL provides arithmetic, comparison, logical, and aggregation operators for filtering and manipulating the metrics data (scalars, instant vectors, or combinations of both)

- Function: take instant vectors (or scalars and range vectors in some cases) as input, transform and analyze time series data, and return a new value, usually another instant vector

# Expression Examples

- Calculating the percentage of memory used on a system

```
(1 - node_memory_MemFree_bytes / node_memory_MemTotal_bytes)*100
```

`node_memory_MemTotal_bytes` = Total physical memory in bytes (scalar)

`node_memory_MemFree_bytes` = Free memory in bytes (instant vector)

# Expression Examples

- Calculating the average cpu utilization of 1-minute range

```
100 - (avg(rate(node_cpu_seconds_total{mode="idle"}[1m])) * 100)
```

`node_cpu_seconds_total` = Total CPU seconds in that specific mode

`node_cpu_seconds_total{mode="idle"}[1m]` = range vectors of 4 values each (config to measure every 15 seconds, 1 min = 4 values)

**rate(v range-vector)** Used for counter, calculates the per-second average rate of increase of the time series in the range vector v (return instant vector)

**avg(v)** divides the sum of v by the number of aggregated samples in the same way as the / binary operator

# Aggregation Operators

- PromQL provides several aggregation operators for combining related time series (similar to groupby in pandas)

| Operator | Description |
|---|---|
| sum | Calculates the sum of values across time-series. |
| avg | Computes the average of values across time-series. |
| min | Finds the minimum value across time-series. |
| max | Finds the maximum value across time-series. |
| count | Counts the number of time-series. |
| stddev | Calculates the standard deviation across time-series. |
| stdvar | Calculates the variance across time-series. |
| quantile | Computes a specific quantile (e.g., 0.95). |
| topk | Selects the top k time-series by value. |
| bottomk | Selects the bottom k time-series by value. |

# Aggregation Dimension Control

- Aggregation operator allows the use of “without” and “by” to exclude or include labels in the calculation

```
sum(node_cpu_seconds_total)
sum(node_cpu_seconds_total) without (cpu)
sum(node_cpu_seconds_total) by (cpu)
sum by (cpu) (node_cpu_seconds_total)
```

# Query Prometheus with Python

- We can retrieve data from Prometheus server via Prometheus http api using prometheus_api_client python package.  To use this package, we must install the following packages:

```
pip install prometheus_api_client
pip install pandas
```

- prometheus_api_client provides several functions.  See example in the jupyter notebook for more details

prometheus

prometheus.ipynb prometheus U

prometheus.ipynb > Retrieving Data from Prometheus > We first import all necessary packages and connect to our local prometheus server

Generate + Code + Markdown | Run All | Clear All Outputs | Outline ··· Select Kernel

# Retrieving Data from Prometheus

We can retrieve data from Prometheus server via Prometheus http api using prometheus_api_client python package. To use this package, we must install the following packages:

```
pip install prometheus_api_client
pip install pandas
```

We first import all necessary packages and connect to our local prometheus server.

```python
from prometheus_api_client import PrometheusConnect,  MetricSnapshotDataFra
from prometheus_api_client.utils import parse_datetime
from datetime import timedelta
```

Python

```python
prom_url = "http://localhost:9090"
prom = PrometheusConnect(url=prom_url, disable_ssl=True)
```

Python

main* Launchpad 0 0 docker-desktop default Update is ready, click to restart. { } Cell 2 of 17

# Conclusion

- Selecting proper datastore that suits to data characteristics is very important
- Time Series Database provides several important features for analyzing and handling time series data
- Prometheus is a very TSDB, especially for observability
- PromQL is Prometheus Query Language supporting data model with metrics and labels, built-in aggregation operators, and comprehensive time series functions

# Recommended Resources

- https://medium.com/kbtg-life/observability-pt-1-ทำความรู้จักกับ-observability-กันเถอะ-5b2561bc4365
- https://wbassler23.medium.com/getting-started-with-prometheus-pt-1-8f95eef417ed
- https://betterstack.com/community/guides/monitoring/prometheus-metrics-explained/
- https://medium.com/@snehatuladhar098/beginners-hands-on-guide-monitor-your-computer-with-prometheus-and-grafana-e3c9011e47c8
- https://pypi.org/project/prometheus-api-client/
- https://promlabs.com/promql-cheat-sheet/